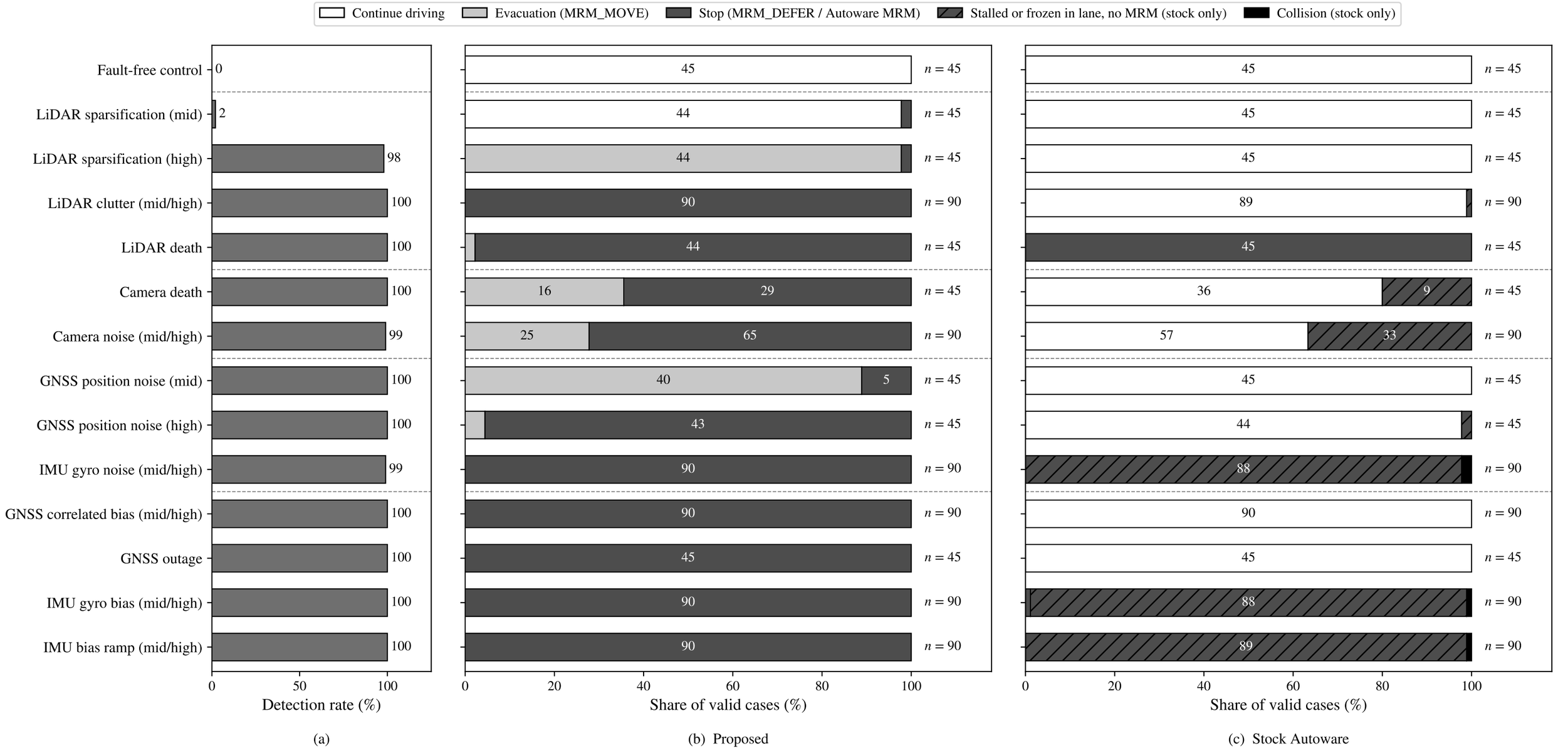
Continue driving
Evacuation (MRM_MOVE)
Stop (MRM_DEFER / Autoware MRM)
Stalled or frozen in lane, no MRM (stock only)
Collision (stock only)
Fault-free control
LiDAR sparsification (mid)
LiDAR sparsification (high)
LiDAR clutter (mid/high)
LiDAR death
Camera death
Camera noise (mid/high)
GNSS position noise (mid)
GNSS position noise (high)
IMU gyro noise (mid/high)
GNSS correlated bias (mid/high)
GNSS outage
IMU gyro bias (mid/high)
IMU bias ramp (mid/high)
0
2
98
100
100
100
99
100
100
99
100
100
100
100
45
44
44
90
44
16
29
25
65
40
5
43
90
90
45
90
90
45
45
45
89
45
36
9
57
33
45
44
88
90
45
88
89
n = 45
n = 90
0
50
100
Detection rate (%)
0
20
40
60
80
100
Share of valid cases (%)
(a)
(b) Proposed
(c) Stock Autoware